Rithvik Santosh Shetty

Python Developer | Backend, REST APIs, Applied ML
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Summary

Python developer with a B.E. in Artificial Intelligence and Machine Learning. Builds **Python** backends and **REST APIs** (**FastAPI**, Node.js) on **PostgreSQL**, **Redis** and Supabase, with **Docker** for deployment. Shipped a YOLOv8 vision pipeline into a live industrial workflow during a Python internship, and built a multilingual voice assistant that placed Top 12 at Sarvam AI's national hackathon. Daily user of **Git**, **GitHub** and Linux, and quick to pick up new frameworks.

Technical Skills

- **Python: Python** (**OOP**, **data structures**, **exception handling**), **FastAPI**, **Pandas**, **NumPy**, Jupyter
- **APIs and Backend: REST APIs**, **FastAPI**, Node.js, **API integrations** (Twilio, WhatsApp, Razorpay, Stripe, Sarvam AI), Cloudflare Workers
- **Databases: SQL**, **PostgreSQL**, Supabase, **Redis**/Upstash (**NoSQL** key-value)
- **Tools and Cloud: Git**, **GitHub**, **Docker**, Linux, Vercel, **AWS** (Introduction to Machine Learning on AWS)
- **ML and AI:** PyTorch, scikit-learn, XGBoost, YOLOv8, OpenCV, Claude API, Gemini API, Model Context Protocol, LangChain, **RAG** (Pinecone, BM25, Cohere Rerank)
- **Frontend:** TypeScript, JavaScript, React, Next.js, React Native, Streamlit

Experience

AI Training Specialist Intern, Autonex AI | Powai, Mumbai, India Nov 2026 – Present

- Label data used for AI model training and evaluation, following labeling guidelines to keep annotations accurate and consistent.
- Check labeled datasets to confirm all data is correctly labeled before it is used for training or evaluation.

Python Developer Intern, Ion Chemicals | Navi Mumbai, India Jan 2025 – Apr 2025

- Built a real-time computer vision pipeline in **Python** with YOLOv8 and OpenCV that detects gas leaks and anomalies on industrial pipelines, reducing manual inspection time by 40%.
- Integrated the detector into existing operational workflows with the plant engineering team.

Salesforce and Marketing Automation Intern, Lupin Ltd | Mumbai, India Dec 2025 – Feb 2026

- Automated lead nurturing and outreach workflows in Salesforce CRM, reducing manual follow-up effort and improving campaign response tracking across multiple product lines.
- Built and optimized marketing automation sequences; analyzed campaign metrics (open rates, CTR, conversion) to surface insights for the marketing team.
- Worked with sales and analytics teams to align CRM data pipelines with business reporting requirements.

Freelance Full-Stack Developer, TRCT (trct.in), The Run Club Thane | Remote Mar 2024 – Present

- Sole developer on the club's production Next.js and TypeScript platform for a 4,000+ member community: authentication, **REST APIs**, and **API integrations** for a Razorpay and Stripe e-commerce module. 100 Lighthouse score.

Projects

Hybrid-Search RAG over Internal Docs | Python, FastAPI, Pinecone, LangChain, Claude API Oct 2026

- Built a **FastAPI** RAG service over internal documents with **POST /ingest** and **POST /query** endpoints, returning an answer with citations, a verification flag and per-stage timings.
- Hybrid retrieval: dense search on Pinecone embeddings and BM25 keyword search, merged with Reciprocal Rank Fusion, then Cohere Rerank on top candidates.
- Claude answers from retrieved context only and cites chunks inline; a verification step checks each claim against its cited chunk and flags unsupported ones.
  Code: github.com/rithvikshettyy/Hybrid-Search-RAG-over-Internal-Docs

Praman | Python, FastAPI, Sarvam AI, Twilio WhatsApp API 2026

- Built a **Python** and **FastAPI** backend for a WhatsApp banking and loan assistant for low-literacy rural users: voice or text in and out, in the user's own language, on Sarvam AI speech-to-text, translation and text-to-speech across 22 Indian languages (**API integration**).
- Added a document explainer for scanned bank letters and loan agreements, plus an operator dashboard. Top 12 of ∼100 teams (∼6,000 registrations), Sarvam Buildin' Hours.

- Rebuilt it for health insurance claims at the Paytm Build for India AI Hackathon: claim documents and voice notes on WhatsApp in 11 Indian languages, with agents handling synced threads from a dashboard.
  Code: github.com/rithvikshettyy/praman-paytm

SeaGuard | React Native, FastAPI, Docker, Twilio Sep 2025

- Safety app for fishermen with a **FastAPI** backend: real-time GPS boundary monitoring, weather alerts and automated SMS alerts. Frontend and backend containerized with **Docker** for deployment.
  Code: github.com/rithvikshettyy/SeaGuard

AutoPharmaX | Python, PyTorch, XGBoost, scikit-learn, Streamlit Jul 2025

- Built a multitask deep learning model in PyTorch with dual MLP and graph neural network encoders to predict cancer drug response and drug-pair synergy, benchmarked against XGBoost and scikit-learn ensembles.
- Added agentic tooling to query drug-efficacy data and a chatbot that turns model outputs into plain-language guidance.
  Code: github.com/rithvikshettyy/autopharmax
- Wrote up the method and results as a conference paper with the team; reviewers returned a revise-and-resubmit.

AnyFormat | Next.js, TypeScript, Upstash Redis, BullMQ, Razorpay 2026

- Self-hosted file conversion SaaS: queued jobs over Sharp, MuPDF, FFmpeg, LibreOffice and Tesseract OCR, with auth, payments (**API integration**) and bot protection.
  Live: anyformat.rithvikshetty.in

Education

B.E. in AI and Machine Learning, SIES Graduate School of Technology (Univ. of Mumbai) Nov 2022 – May 2026

- Coursework: Machine Learning, Deep Learning, NLP, Image and Video Processing, Advanced AI, Big Data Analytics, Data Analytics and Visualization

Achievements and Certifications

- Top 12 of ∼100 on-site teams (∼6,000 registrations), Sarvam Buildin' Hours hackathon by Sarvam AI, Bengaluru (2026)
- Selected among ∼100 finalist teams from 6,000+ registrations, Paytm Build for India AI Hackathon, Mumbai (2026)
- Most Product Readiness Award, HackCelestial 2.0 (200+ teams), Pillai College of Engineering (2025)
- Award for SeaGuard presented by the Director, ISRO (2025)
- Introduction to Machine Learning on AWS (AWS and Coursera, 2025)
- Introduction to Model Context Protocol (Anthropic, 2025)

Positions of Responsibility

- **Secretary, Technical Team, SIES GST** (2025 – 2026): Led ByteCamp'25/26, a national-level hackathon with 2,400+ registrations and 40 finalists; coordinated technical events, workshops and mentorship for the college tech community.
- **Design Lead, TEDxSIESGST** (2025 – 2026): Directed the design team on promotional graphics, stage visuals and communication assets.
- **Social Media Lead, E-Cell, SIES GST** (2024 – 2025): Managed social media across platforms to drive engagement and event visibility; coordinated promotional campaigns.